

Algebraic Techniques for Meson Masses and Decays for the Unquenched Quark Model

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Novel algebraic methods are applied to quarkonium spectrum in the unquenched quark model, which incorporates the coupling between $Q\bar{Q}$ and $(Q\bar{q})(q\bar{Q})$ degrees of freedom. Working in the discrete position basis, the calculation of matrix elements simplifies considerably in comparison to other approaches, leading not only to improvements in computational efficiency, but also more flexibility in constructing the model and analysing its output. We illustrate several methods for the extraction of hadron parameters including masses, rms radii and phase shifts.

Introduction

Approximate solutions to the Schrodinger equation,

$$\hat{H}|\psi\rangle = E|\psi\rangle,$$

and related problems can be found by expanding the wavefunction in the basis of orthonormal functions,

$$|\psi\rangle = \sum_{n=0}^{N-1} |n\rangle \langle n|\psi\rangle.$$

This constructs an eigenvalue/eigenvector equation for a $N \times N$ matrix whose elements are integrals,

$$\langle m|\hat{H}|n\rangle = \int \langle m|x\rangle \hat{H} \langle x|n\rangle dx,$$

classical orthogonal polynomials are commonly used for $\langle x|n\rangle$, with a suitable choice of polynomial determined by the domain of the position variable x of the problem.

- Hermite Polynomials ($-\infty < x < \infty$)
- Associated Laguerre Polynomials ($0 \leq x < \infty$)
- Gegenbauer Polynomials ($-1 \leq x \leq 1$)

The approximation is exact in the limit $N \rightarrow \infty$, so higher matrix sizes are essential. The problem involves solving integrals, which is numerically intensive, but using algebraic techniques which find the solutions to these integrals high matrix sizes are computationally efficient and numerically stable.

The Discrete Position Basis

The new approach involves expanding into a different set of basis functions,

$$|\psi\rangle = \sum_{i=1}^N |i\rangle \langle i|\psi\rangle,$$

which are related by an orthogonal transformation to the classical orthogonal polynomials,

$$|i\rangle = \frac{1}{\mathcal{N}_i} \sum_{n=0}^{N-1} |n\rangle \langle n|x_i\rangle,$$

where $\mathcal{N}_i$ is a normalisation factor and the coefficient $\langle n|x_i\rangle$ is the n th polynomial evaluated at the i th root of the N th polynomial. This choice as a result of the three term recurrence relation,

$$\hat{x}|n\rangle = c_{n+1}|n+1\rangle + b_n|n\rangle + c_n|n-1\rangle,$$

implies that in this new basis the operator $\hat{x}$ is diagonal,

$$\langle i|\hat{x}|j\rangle = x_i \delta_{ij}.$$

Which provides the advantage that for large N the matrix of an arbitrary function of $\hat{x}$ is also diagonal,

$$\langle i|f(\hat{x})|j\rangle = f(x_i) \delta_{ij},$$

which reduces the number of matrix elements to calculate to N as opposed to the original N^2 and the matrix elements are simply $f(x_i)$. The kinetic term is the only part of the Hamiltonian which is not diagonal however those terms can also be obtained algebraically so in total there is no integration required at any point.

Quenched Quark Model

To illustrate the technique, consider the quark potential model for $b\bar{b}$ mesons,

$$\hat{H} = \frac{\hat{q}^2}{m_b} - \frac{4\alpha_s}{3r} + \lambda r + C,$$

which when transformed into dimensionless variables x and p defined by $x = r\beta$ and $p = \frac{q}{\beta}$ is,

$$\hat{H} = \frac{\beta^2}{m_b} \hat{p}^2 - \frac{4\alpha_s \beta}{3} \frac{1}{x} + \frac{\lambda}{\beta} x + C.$$

For a given orbital angular momentum, the matrix of $\hat{H}$ is constructed in the basis of Associated Laguerre polynomials, building the matrices of p^2 , $1/x$ and x algebraically as described. The meson masses are then the eigenvalues of $\hat{H}$ which are stable under variation of β and the meson wavefunctions are given by the eigenvectors, $\langle i|\psi\rangle$, for example the position space wavefunctions are

$$\langle x|\psi\rangle = \sum_{i=1}^N \langle x|i\rangle \langle i|\psi\rangle, \quad \text{with } \langle x|i\rangle = \frac{c_N \langle x|N\rangle \langle N-1|x_i\rangle}{\mathcal{N}_i x - x_i},$$

and the expectation values of operators such as the rms radii are matrix products,

$$\langle \mathcal{O} \rangle = \langle \psi|\mathcal{O}|\psi\rangle = \sum_{i,j=1}^N \langle \psi|i\rangle \langle i|\mathcal{O}|j\rangle \langle j|\psi\rangle.$$

Unquenched Quark Model

By extending the matrix from $N \times N$ to $2N \times 2N$ the model can incorporate light-quark pair creation $b\bar{b} \rightarrow (b\bar{q})(q\bar{b})$, which causes mass shifts, and for states above threshold, strong decay. In the heavy-quark limit (ignoring spin-dependent splittings), the sum over couplings to the related channels $B\bar{B}$, $B\bar{B}^*$, $B^*\bar{B}$ and $B^*\bar{B}^*$ gives the same mass shift and decay width for all states in a single multiplet (such as 1P_1 , 3P_0 , 3P_1 and 3P_2 [1] [2]). Hence for simplicity consider a single " $B\bar{B}$ " channel, effectively an average of the physical channels.

$$\begin{array}{c} b \\ \bar{b} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} \begin{array}{c} b \\ q \\ q \\ \bar{b} \end{array} \longleftrightarrow \begin{array}{c} b\bar{b} \\ B\bar{B} \end{array} \begin{pmatrix} H_1 & U(x) \\ U(x) & H_2 \end{pmatrix}$$

The model Hamiltonian includes a part H_1 , equal to the quenched quark model above, in addition to

$$\hat{H}_2 = \frac{\beta^2}{m_B} \hat{p}^2 + V(x),$$

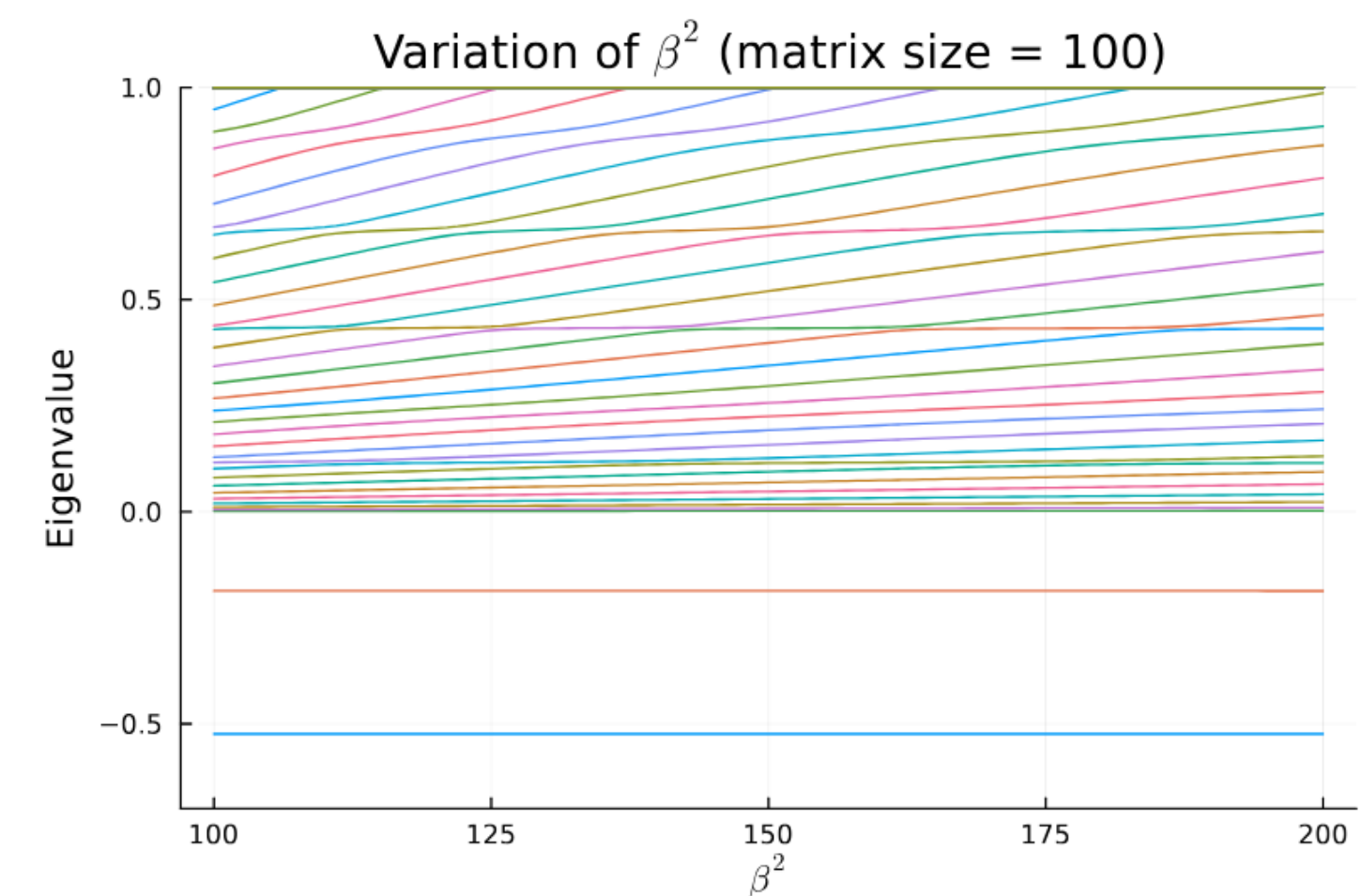
for the $B\bar{B}$ channel, and $U(x)$ for the coupling between the two degrees of freedom from [3],

$$U(x) = \frac{\gamma\sqrt{\pi}}{8} \frac{x}{\beta R^2} e^{-\frac{x^2}{4\beta^2 R^2}}$$

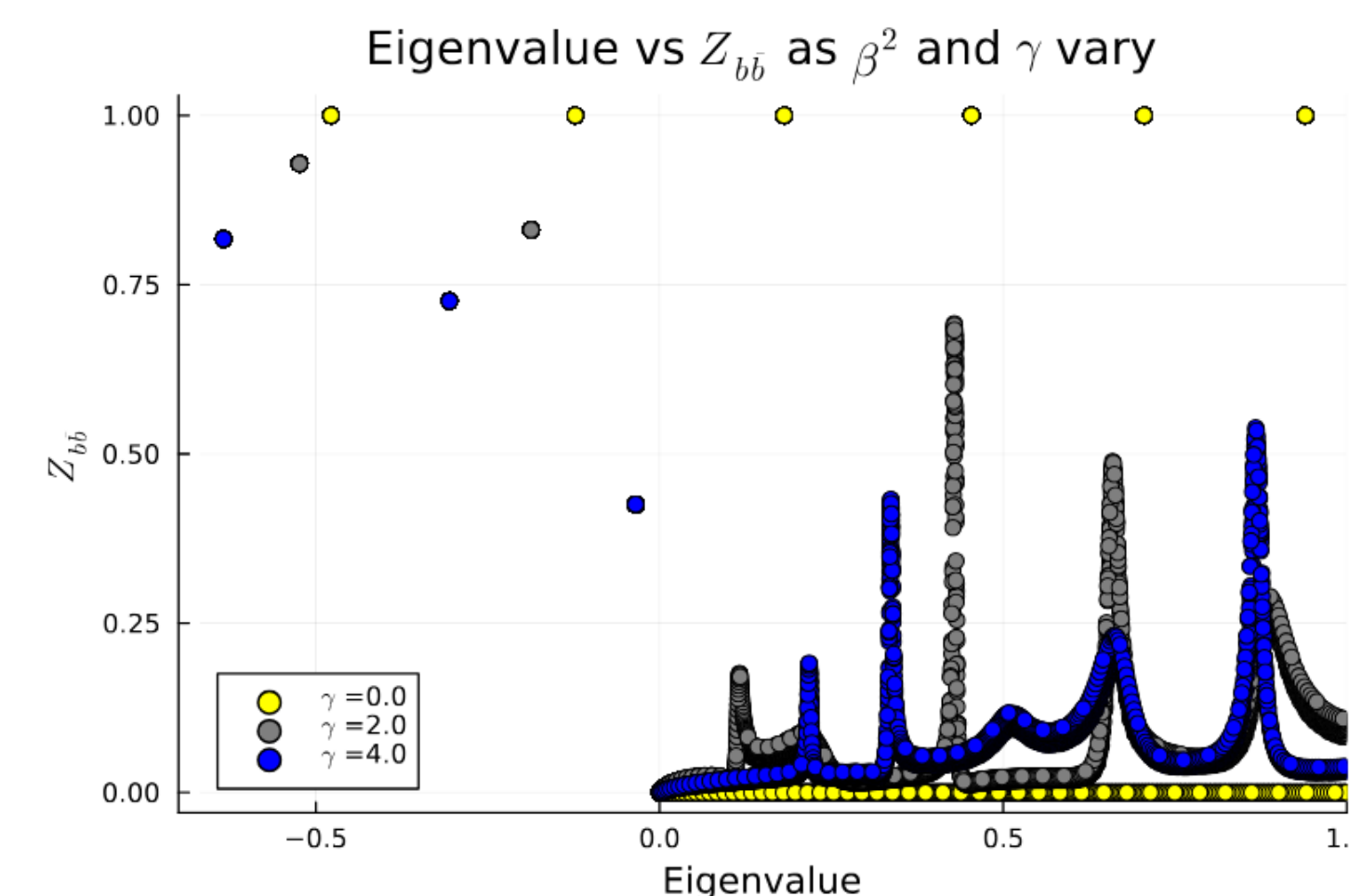
and $V(x) = -g\beta \frac{e^{-\frac{m\pi x}{\beta}}}{x}$ which is the potential in the " $B\bar{B}$ " channel; a Yukawa potential representing the possibility of pion exchange between the $B\bar{B}$ mesons.

Results of the Unquenched Quark Model

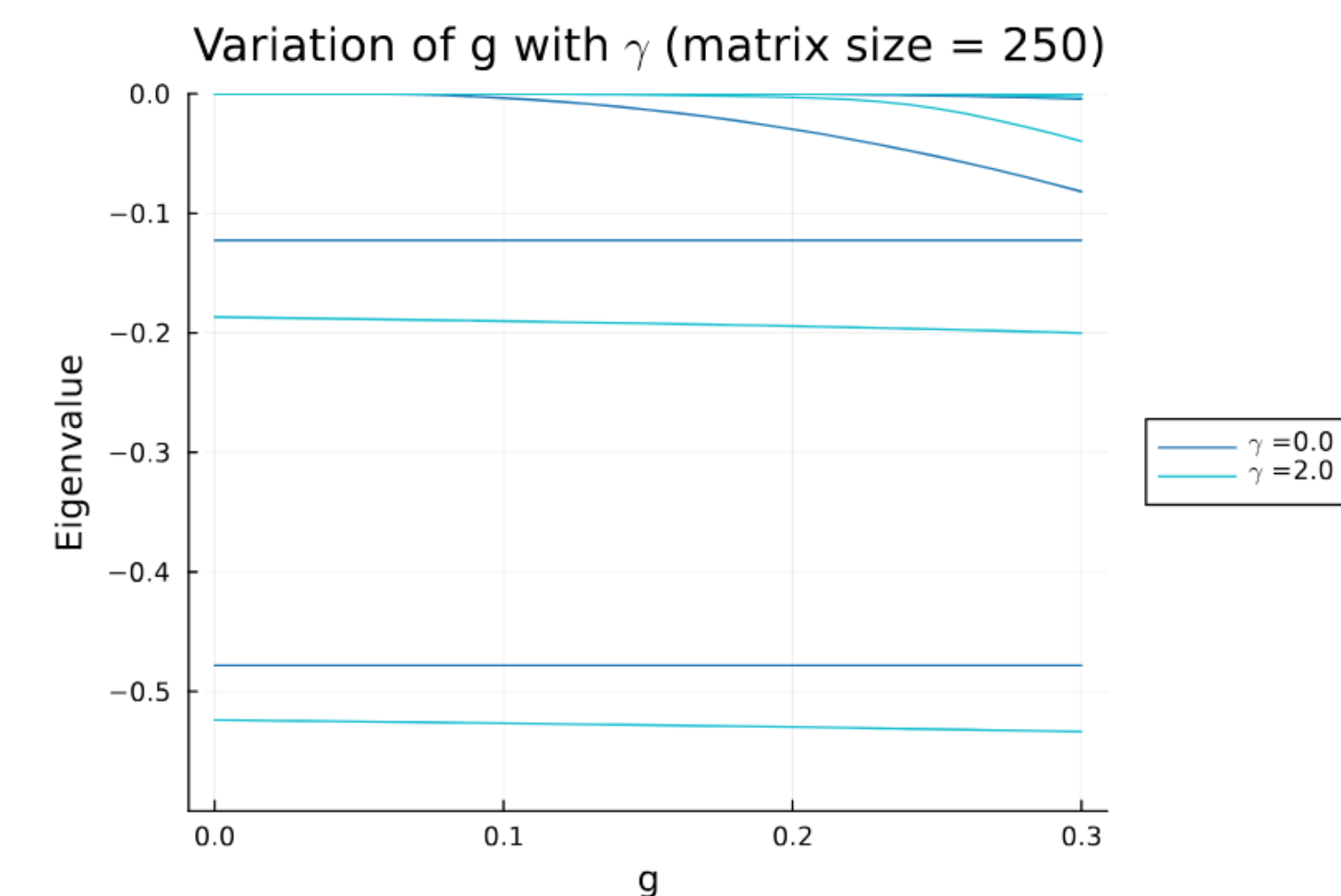
Firstly, consider H_1 and H_2 with all coupling and pion exchange potentials turned off, there is no mixing between the two channels, then turn on the coupling and vary its strength using γ . For illustration consider $b\bar{b}$ with $l = 1$ and $B\bar{B}$ with $l = 0$. The Hamiltonian was shifted so that 0.0 on plots represents the $B\bar{B}$ threshold. Below the threshold there are bound states but above the threshold the continuum spectrum is mixed with resonances characterised by avoided level crossings in a stabilisation plot.



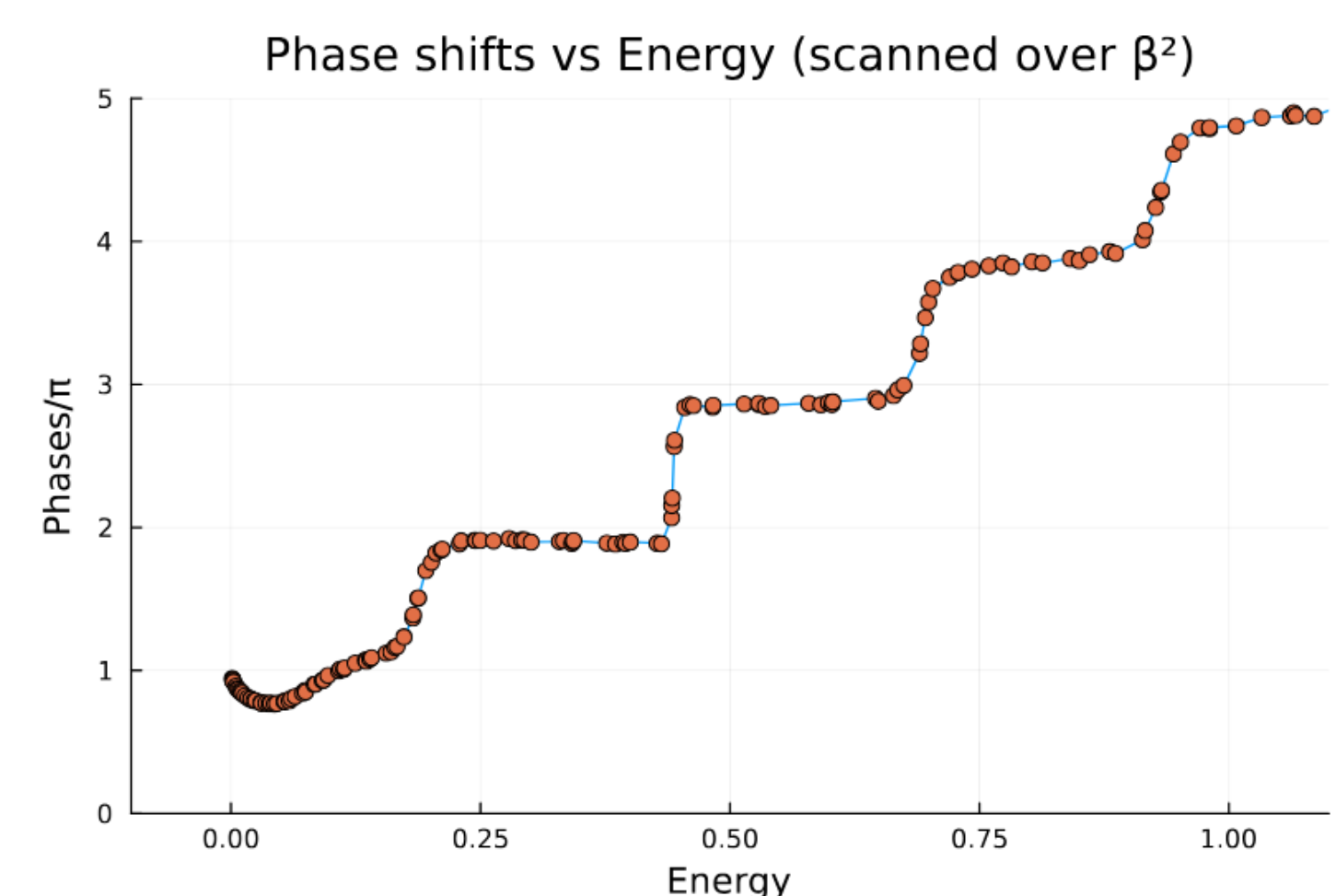
A useful quantity for understanding the physics is $Z_{b\bar{b}}$ the proportion of $b\bar{b}$ in the final eigenvector composition. When varying the coupling, mixing between the states occurs and the plots of $Z_{b\bar{b}}$ give a clear visual resonance signal, with broader resonances having a larger energy spread. The bound states below the threshold remain bound and as the energy eigenvalue approaches the threshold $Z_{b\bar{b}}$ decreases until the energy crosses the threshold and resonance states are present centred around the resonance energy. The energies of the states deepen as γ is increased and the generation of an additional bound state below threshold can be caused, which consists of the largely $B\bar{B}$ channel, similar results are found by [4].



Finally, the effect of turning on the pion exchange with variable g is to deepen the bound states until additional bound states emerge and become more bound as g grows.



Another method for characterising the resonances is to study the phase shifts of the $B\bar{B}$ component of the wavefunction, which at large distance behave approximately as free particles. In a plot of the phase shifts against the energy a transition of π in the phase shift occurs in the presence of a resonance. The resonance energy can be determined from the point of inflection, and the decay width by fitting to $\delta = \delta_0 - \tan^{-1}(\frac{\Gamma/2}{E-E_R})$. For $\gamma = 1.0$ there is a mix of broader and narrower resonance states.



Conclusion

Using a discrete position basis as an expansion of the wavefunction is a valid and computationally efficient method of studying the unquenched quark model and finding meson masses and decays as well as their wavefunctions providing the extraction of hadron parameters such as the rms radii. The model can be made more realistic by extending to multiple channels and using the discrete position basis this extension is very easy as an extension of N extra rows and columns in the combined matrix leads to only N extra matrix elements rather than N^2 meaning accessing very large matrices is plausible. Complex scaling these matrices with $x \rightarrow xe^{i\theta}$ and $p \rightarrow pe^{-i\theta}$ to find resonance properties in the complex plane would provide another method of finding resonances in these models.

References

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